Identification and Re-analysis of Undiagnosed Heritable Connective Tissue Disorders within the Center for Individualized Medicine Diagnostic Odyssey Cohort

Rhiannon Edwards[1], Joel A. Morales-Rosado[2], and Eric W. Klee[2,3]

Summer Undergraduate Research Fellowship Program, Mayo Graduate School of Biomedical Sciences, Mayo Clinic, Rochester, Minnesota, USA

[2]Center for Individualized Medicine, Mayo Clinic, Rochester, Minnesota, USA, [3]Department of Clinical Genomics, Mayo Clinic, Rochester, Minnesota, USA

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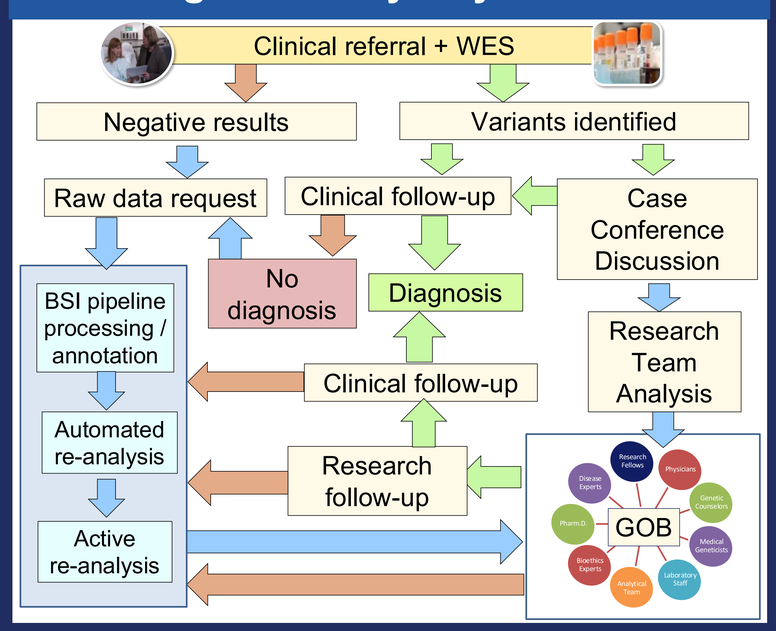
Introduction

- **Connective tissue diseases** are any condition where the structural connections between body parts are affected. Non-heritable disorders affect up to 5% of the world population.
  - **Autoimmune:** systemic lupus erythematosus, rheumatoid arthritis, scleroderma, ankylosing spondylitis, and Sjögren's syndrome
  - **Environmental or unknown causes**: scurvy and other vitamin deficiencies, drug-induced diseases, fibromyalgia, benign and malignant tumors, osteomyelitis, and certain infections
- **There are 200 recognized heritable connective tissue disorders**
  - **Overall incidence** is largely unknown, but estimated to be 1/5000
  - **Many cases are monogenic**; however, **polygenic** pathogenesis is implicated in some disorders, such as the Ehlers-Danlos Syndromes.
  - **Wide variety of phenotypes** with some degree of system involvement
  - **Gene panels** offered by all major genetic testing companies, ranging from 15 to 67 genes covered, and cost from $1000-$3000

Diagnostic Odyssey Process



Heritable Connective Tissue Disorders

- Result from genetic variations that **disrupt the formation, organization, and maintenance of the extracellular matrix**
- **Connective tissues** include the tendons, ligaments, cornea, organ membranes, cartilage, muscle matrix, bones matrix,, and fascia
- **Major components**: elastin, collagen, fibrin, fibronectin, laminin, tenascin, myosin, fibrillin, and proteoglycans
- **Major disorders** include Marfan Syndrome, Ehlers-Danlos Syndromes, Osteogenesis imperfecta, oculoskeletal dysplasias, Alport Syndrome, cutis laxa, and Bethlem myopathy

Typical Diagnostic Process

Clinical Exam:
- History
- Physical Exam
- Beighton Score
- Skin Exam
- Vitals and orthostatics
- Neurological exam

Clinical Testing:
- EKG and Echocardiogram
- Blood work

If warranted:
- EMG and MRI
- Skin biopsy
- Tilt table test
- X-rays/imaging

Genetic Testing:
- Connective tissue panel
- Family Testing

If warranted:
- Whole Exome Sequencing
- RNA and Sanger sequencing

Common Exam Findings

Hypermobility, myalgia, myopathy or neuropathy, frequent dislocations, immune dysfunction, soft, stretchy skin, scoliosis, osteoarthritis, autonomic dysfunction, poor wound healing, hearing loss, ectopia lentis, chronic pain

Aim

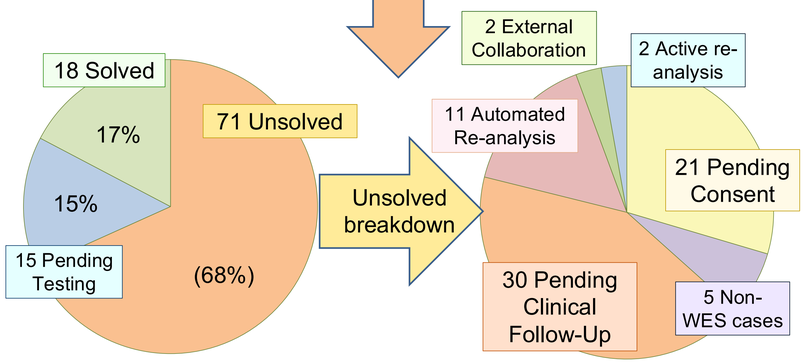
Identify and re-analyze unsolved connective tissue disorder cases using Whole Exome Sequencing (WES) data from the Center for Individualized Medicine Diagnostic Odyssey Cohort

Cohort Analysis

2,009 total cases (as of July 2019)

Phenotyping

102 cases (5.1%) identified as suspected connective tissue disorders



Re-Analysis Methods

Table 2A. Inclusion Criteria
- Spontaneous arterial dissection
- Frequent joint dislocation
- Marfanoid habitus
- Cigarette-paper scarring
- Spontaneous organ rupture
- Clinically suspected
- Repeated fractures
- Mitral valve prolapse
- Spontaneous pneumothorax
- Delayed wound healing
- Soft, doughy skin
- Delicate, hyperextensible skin
- Pelvic floor prolapse
- Joint hypermobility
- Early-onset osteoarthritis
- Dysautonomia
- Scoliosis or kyphosis
- Pectus deformity

Phase 1: Phenotyping and Identification of Cases

PHENOTIPS

Table 2A and 2B: Inclusion and Exclusion criteria for identifying likely cases of connective tissue disorders. Frequently, patients were referred with either a diagnosis or clinical suspicion of a connective tissue disorder.

Phase 2: WES Raw Data Re-Analysis

Table 2B. Exclusion Criteria
- Known mitochondrial disease
- Metabolic disorder
- Intractable epilepsy
- Porphyria
- Developmental regression
- Parkinsonism
- Huntington's chorea
- Endocrine disorder
- Myelopathy
- Liver abnormalities
- Hemolytic anemia
- Inflammatory bowel disease
- Spina bifida
- Hyperlipidemia
- Extreme obesity
- Microcephaly
- Craniosynostosis

Emedgene Variant Curation: <1% population frequency, matches expected inheritance, High quality coverage, High impact variant

- Strategies vary based on inheritance and available family testing-

Proband or Duo:
1) Most likely by AI
2) Candidates by AI
3) Hom in AR conditions
4) Two VUS in one gene
5) X-linked conditions
6) Het for AD conditions
7) Het for AR conditions
8) GUS

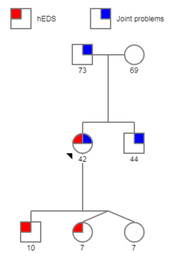
Trio or multiples:
1) Most likely by AI
2) Candidates by AI
3) De Novo
4) Hom in AR condition
5) Compound Het
6) X-linked conditions
7) Het for AD conditions
8) Het for AR conditions
9) GUS

VUS or GUS Identified: Phenotypic overlap, population info, tissue expression, variant impact, genotype overlap, disease mechanism

Main Findings

Periodical WES re-analysis has the potential to identify new gene-disease associations and lead to the diagnosis of unsolved connective tissue disorder cases

Re-Analysis of Case #974



Phenotype:
- Hypermobile EDS
- Muscle weakness
- Early osteoarthritis
- Neuropathy

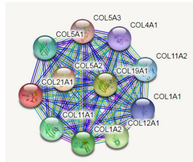
Previous Genetic Testing:
- Clinical WES and mitochondrial DNA: negative

Candidate Found:

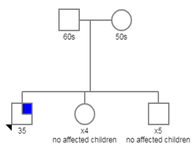
COL9A3 (het) c.338 G>A p.Gly130Ser (maternal)

Disease: Multiple epiphyseal dysplasia, with or without myopathy
gnomAD: 0.083% (no homozygotes)
ClinVar and HGMD: VUS/not reported
In silico (SIFT/Polyphen-2/Mutation Taster/CADD/PredictSNP2): **DAMAGING** Deleterious / Probably Damaging/ Disease Causing/ 27.7 / neutral
Location: Chr20(GRCh37): g. 61453127 G>A | Exon 8 of 32
Domain: Gly-X-Y repeat, collagen IX trimer, collagen helix repeat region
Function: extracellular matrix organization, most highly expressed in the tibia (tibial nerve, surrounding tissues).
Inheritance: maternal; unknown if mother has phenotype

- Could explain myopathic symptoms, arthritis, and pain
- Additional finding: COL21A1 (GUS), interacts with other collagens
- **Next steps**: sequencing family members, epiphyseal x-rays, and fibroblast collagen staining



Re-Analysis of Case #927



Phenotype:
- 30s male
- Congenital lower limb contractures
- Spastic gate,
- Hip osteoarthritis
- Collagen VI staining negative

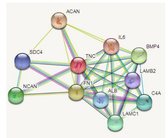
Previous genetic testing:
- Duo WES results determined benign

Candidate Found:

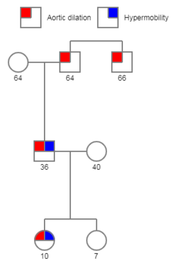
TNC (het) c.3481 T>C p.Ser1161Pro

Disease: Sensorioneuronal hearing loss with muscle weakness (2 families)
gnomAD: not reported
ClinVar and HGMD: not reported
In silico (SIFT/Polyphen-2/Mutation Taster/CADD/PredictSNP2): **CONFLICTING** Tolerated / Probably Damaging/ Polymorphism/ 24.1 / neutral
Location: Chr9(GRCh37): 117826932 A > G | Exon 11 of 28
Domain: Fibronectin type-III repeat 6, fibrin/fibronectin cross-linking region
Function: collagen fibril organization, maintaining fibril interface , wound repair
Inheritance: not maternal, father unknown

- Abnormal TNC in degenerating human tendons
- Tenascin-deficient: myopathy, similar biopsy
- **Next Steps**: Requesting paternal and sibling genetic testing, auditory testing, and muscle biopsy staining for TNC, FBN, LAM



Re-Analysis of Case #755



Phenotype:
- Aortic root dilation
- Multiple thrombotic events, stroke
- Mild marfanoid features
- Hypermobility

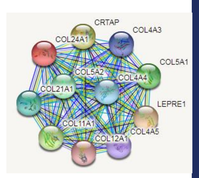
Previous genetic testing:
- Marfan gene panel and WES: two VUS, both benign

Candidate Found:

THBS3 (het) c.1559 G>A p.Arg520Gln (paternal)

Disease: not known (GUS)
gnomAD: 0.03% (no homozygotes)
ClinVar and HGMD: not reported
In silico (SIFT/Polyphen-2/Mutation Taster/CADD/PredictSNP2): **DAMAGING** Tolerated / Probably Damaging/ Disease causing/24.1/ Deleterious
Location: Chr1(GRCh37): 155170391 C > T | Exon 14 of 23
Domain: thrombospondin type-III repeat 2, disulfide-bond region
Function: Cardiovascular development and repair; interacts with fibrinogen, fibronectin, laminin and type V collagen
Inheritance: paternal, strong family history

- THBS1 and THBS4 variants associated with higher risk of myocardial infarction
- Role in arterial development and repair; also interacts with fibrin and fibrillin
- Second GUS: COL24A1, regulates COL1A1
- **Next Steps:** Requesting testing for other family members and fibroblast staining for collagen I, III, and V structure



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